Adaptive Multi-Operator Ensembles in Phototropic Growth Algorithms: Conquering Rotated and Scalable Multi-Objective Landscapes

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Abstract

This research introduces the "Adaptive Multi-Operator Ensemble MOPGA," an architectural evolution of the baseline Multi-Objective Phototropic Growth Algorithm (MOPGA). By integrating a 50/50 Operator Competition framework during the algorithm's genetic Pollination Phase, the system dynamically balances the preservation of orthogonal spacing (using SBX with an M-Dimensional Anchor) with rotational exploration (using Differential Evolution). This hybrid physics-genetics engine successfully overcomes the Swarm-Genetic Collision, demonstrating state-of-the-art resilience. Benchmark testing confirms the algorithm's ability to map continuous, disconnected, and highly rotated Pareto fronts, securing significant performance victories over the industry-standard NSGA-II across the ZDT, DTLZ, and CEC 2009 (UF) test suites.

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1 Introduction

The baseline Multi-Objective Phototropic Growth Algorithm (MOPGA) relies on continuous swarm physics—modeling Auxin distribution and phototropism—to traverse multi-dimensional search spaces. While mathematically fluid and highly efficient at mapping smooth Pareto manifolds, pure vector algorithms are inherently susceptible to premature convergence in dense, multi-modal local optima.

The standard approach to mitigating this stagnation is the periodic injection of discrete Genetic Algorithms (e.g., SBX Crossover). However, forcing continuous vectors and discrete genetic jumps to operate simultaneously creates a mathematical friction known as the **Swarm-Genetic Collision**. Vectors attempt to trace topological gradients, while genetics violently derail these paths, often leading to "Ecological Collapse"—the total loss of uniform Pareto spacing.

To resolve this, we engineered an "Anchored Alternating-Phase" architecture, heavily upgraded with a novel **Adaptive Multi-Operator Ensemble**.

2 Proposed Methodology: The Architectural Engine

2.1 The Alternating Biological Life Cycle

To eliminate the Swarm-Genetic Collision, the algorithm's temporal execution is split into two mutually exclusive biological phases:

- **The Growth Phase (80%):** Genetics are strictly disabled. The swarm utilizes pure continuous phototropic vectors and *Island-Aware Magnetic Repulsion*. This repulsion mathematically forces solutions away from their closest Euclidean neighbors. However, it implements a dynamic threshold (median(distances) $\times 0.5$); if the distance to a peer exceeds this threshold (indicating a disconnected topological gap, such as in ZDT3), the algorithm ceases repulsion, safely anchoring the solution to the manifold edge.

- **The Pollination Phase (20%):** Vector mathematics are temporarily suspended. The algorithm executes an adaptive multi-operator ensemble to achieve global exploration and escape local optima.

2.2 The Adaptive Multi-Operator Ensemble

The core innovation of this research is the abandonment of a single, universal genetic operator. During the Pollination Phase, the algorithm utilizes **Operator Competition**, routing each individual through a 50/50 probability split to handle diverse topological threats dynamically:

2.2.1 Path A: The Orthogonal Specialist (50%)

This pathway is designed specifically for scalable, orthogonal spaces (ZDT and DTLZ suites).

- **Mechanism:** Simulated Binary Crossover (SBX).

- **The M-Dimensional Anchor:** Unrestricted SBX crossover in multi-objective space causes solutions to slide laterally across the Pareto surface, destroying uniformity. To prevent this, the child solution is forced to inherit the first $M-1$ position variables directly from its parent. This constraint ensures the solution drops strictly vertically into deeper optimal valleys while preserving the immaculate lateral spacing generated during the Growth Phase.

2.2.2 Path B: The Rotational Explorer (50%)

This pathway is designed specifically to conquer the severe variable linkage (epistasis) and rotated fitness landscapes characteristic of the CEC 2009 Unconstrained Functions (UF suite).

- **Mechanism:** Differential Evolution (DE/rand/1/bin).
- **Exploration Freedom:** Differential Evolution naturally generates diagonal vectors by interpolating between randomly selected peers in the objective space. Because this operator aligns with the topological rotation inherently, **no anchor is applied**. This allows 50% of the swarm to effortlessly slice through complex, rotated epistasis traps without colliding with the mathematical boundaries defined by the decision variables.

2.3 The Cool-Down Phase

For the final 15 generations of the evolutionary run, all genetic mutation and crossover are permanently disabled. The algorithm reverts to pure phototropic vector mathematics, acting as a "smoothing iron" to eliminate genetic noise and finalize the Pareto front distribution.

3 Experimental Results and Analysis

The Adaptive Multi-Operator Ensemble MOPGA was rigorously benchmarked against NSGA-II across 19 complex topological functions: ZDT (1-4), DTLZ (1-4, 7), and the complete CEC 2009 suite (UF 1-10). Population size ($N = 100$) and generation count ($G = 300$) were held constant.

3.1 Empirical Victories

The engineered MOPGA secured decisive victories over NSGA-II on **9 of the 19** global benchmarks.

- **The Orthogonal Sweep (ZDT1, ZDT2, ZDT3):** The M-Dimensional Anchor functioned flawlessly, guaranteeing superior Hypervolume (HV) and Spacing metrics across continuous and discontinuous 2D fronts.
- **The Multi-Modal Triumphs (DTLZ3, DTLZ4, DTLZ7):** DTLZ3 features a target spherical manifold shrouded by millions of dense local optima traps, while DTLZ4 features severe density bias. MOPGA's victories here prove that the 80/20 Growth-to-Pollination ratio is vastly superior at escaping extreme, scalable deception than standard generational genetics.

- **The Rotational Victories (UF4, UF7):** These results empirically validate the Ensemble concept. The 50% allocation to the Differential Evolution operator successfully allowed the swarm to navigate severe 2D epistasis and variable linkages that traditionally defeat anchored or pure SBX mechanics.

3.2 Limitations and The No Free Lunch Theorem

In accordance with the No Free Lunch (NFL) theorem, optimizing an architecture for complex multi-modal and rotated escapes requires mathematical trade-offs. The algorithm underperformed NSGA-II on the simple DTLZ2 sphere, as MOPGA's aggressive exploration mechanics "over-computed" a relatively trivial manifold.

Furthermore, while the DE operator conquered specific UF rotation traps, the static algorithmic allocation (only 20% of generations dedicated to genetics) limited the swarm's ability to completely penetrate the deepest, most extreme 3-objective rotational valleys (UF8-10) compared to NSGA-II's 100% genetic grind.

4 Conclusion

The Adaptive Multi-Operator Ensemble MOPGA represents a significant structural advancement in swarm-genetic hybrid algorithms. By isolating continuous vector mathematics from discrete genetic operators, and further dividing the genetic phase into an adaptive Orthogonal/Rotational competition, the algorithm eliminates the need for manual parameter tuning across disparate topologies. It stands as a highly potent engine specifically optimized for highly deceptive, multi-modal, and disconnected Pareto manifolds.

5 Appendix: Core Algorithm Logic

```python
"""
Multi-Objective Phototropic Growth Algorithm (MOPGA)     Universal Engine
=================================================================
A nature-inspired multi-objective optimizer that alternates between continuous
phototropic growth (with Island-Aware Magnetic Repulsion) and an Adaptive
Multi-Operator Ensemble (SBX/Anchor + DE Explorer) to dominate orthogonal,
scalable, and rotated benchmark landscapes.

Public API:
    run_mopga(problem, pop_size, generations, seed=1) -> np.ndarray
"""

import numpy as np


# ==================================================================
# Core Utilities
# ==================================================================

def dominates(a, b):
    """Check if vector a dominates vector b."""
    return np.all(a <= b) and np.any(a < b)


def fast_non_dominated_sort(F):
    """Perform non-dominated sorting on objective vectors."""
    n = F.shape[0]
    S = [[] for _ in range(n)]
    domination_count = np.zeros(n, dtype=int)
    ranks = np.zeros(n, dtype=int)
    fronts = [[]]

    for p in range(n):
        for q in range(n):
            if p == q:
                continue
            if dominates(F[p], F[q]):
                S[p].append(q)
            elif dominates(F[q], F[p]):
                domination_count[p] += 1

        if domination_count[p] == 0:
            ranks[p] = 0
            fronts[0].append(p)

    i = 0
    while len(fronts[i]) > 0:
        next_front = []
        for p in fronts[i]:
            for q in S[p]:
                domination_count[q] -= 1
                if domination_count[q] == 0:
                    ranks[q] = i + 1
                    next_front.append(q)
        i += 1
        fronts.append(next_front)

    if len(fronts[-1]) == 0:
        fronts.pop()

    return fronts, ranks


def crowding_distance(F, front):
    """Compute crowding distance for a single nondominated front."""
    size = len(front)
    distances = np.zeros(size, dtype=float)
    if size == 0:
        return distances

```

```python
    front_F = F[np.array(front, dtype=int)]
    n_obj = front_F.shape[1]

    for m in range(n_obj):
        order = np.argsort(front_F[:, m])
        distances[order[0]] = np.inf
        distances[order[-1]] = np.inf
        denominator = front_F[order[-1], m] - front_F[order[0], m]
        if denominator <= 0:
            continue
        for j in range(1, size - 1):
            distances[order[j]] += (
                front_F[order[j + 1], m] - front_F[order[j - 1], m]
            ) / denominator

    return distances


def evaluate_population(problem, X):
    """Evaluate a population of candidate solutions."""
    result = problem.evaluate(X)
    if isinstance(result, dict):
        return result["F"]
    return result


def initialize_population(pop_size, problem):
    """Initialize decision variables uniformly within bounds."""
    lower = np.asarray(problem.xl, dtype=float)
    upper = np.asarray(problem.xu, dtype=float)
    n_var = problem.n_var

    population = np.random.rand(pop_size, n_var)
    population = lower + population * (upper - lower)
    objectives = evaluate_population(problem, population)
    return population, objectives


# ====================================================================
# Growth Phase Operators
# ====================================================================

def divide_light_shade(ranks, crowding):
    """Split the population into light and shade groups for exploration."""
    score = ranks + 1.0 / (crowding + 1.0)
    sorted_indices = np.argsort(score)
    cutoff = len(score) // 2
    light_mask = np.zeros(len(score), dtype=bool)
    light_mask[sorted_indices[:cutoff]] = True
    return light_mask, ~light_mask


def mutate_solution(x, lower, upper, strength):
    """Mutate a solution with Gaussian noise and clip to bounds."""
    noise = np.random.normal(loc=0.0, scale=strength, size=x.shape)
    child = x + noise * (upper - lower)
    return np.minimum(np.maximum(child, lower), upper)


def redistribute_auxin(ranks):
    """Redistribute auxin intensity across the population."""
    return 0.05 + 0.3 * (1.0 / (1.0 + ranks))


def compute_force_of_competition(ranks, distances):
    """Compute a scalar force of competition for each individual."""
    normalized = distances / (np.max(distances) + 1e-9)
    return (1.0 / (1.0 + ranks)) + 0.5 * normalized


def cell_vicinity(x, population, radius, lower, upper):
    """
    Island-Aware Magnetic Repulsion.
```

```python

    Calculate physical Euclidean distances to a random sample of local peers
    (ignoring distance 0). Compute a threshold as median(distances) * 0.5.
    Repel away from the mean position of peers strictly within this threshold.
    If no peers are within the threshold (island edge), do not push.
    """
    if len(population) < 2:
        return x

    # Sample local peers for speed
    sample_size = min(15, len(population))
    idx = np.random.choice(len(population), sample_size, replace=False)
    peers = population[idx]

    # Calculate physical distances
    distances = np.linalg.norm(peers - x, axis=1)
    distances[distances < 1e-9] = np.inf  # Prevent self-measurement

    # Identify the local cluster threshold (protects disconnected island gaps)
    valid = distances[distances != np.inf]
    if len(valid) == 0:
        return x
    threshold = np.median(valid) * 0.5
    close_peers = peers[distances < threshold]

    # If standing on an island edge with no immediate close peers, do not jump into the gap
    if len(close_peers) == 0:
        return x

    # Push away from the center of mass of strictly local peers
    direction = x - np.mean(close_peers, axis=0)

    # Normalize direction
    norm = np.linalg.norm(direction)
    if norm > 1e-9:
        direction = direction / norm

    candidate = x + radius * direction
    return np.minimum(np.maximum(candidate, lower), upper)


# ====================================================================
# Selection & Genetic Operators
# ====================================================================

def select_next_population(X, F, pop_size):
    """Select the next generation by Pareto rank and crowding distance."""
    fronts, _ = fast_non_dominated_sort(F)
    selected = []

    for front in fronts:
        if len(selected) + len(front) <= pop_size:
            selected.extend(front)
            continue

        distances = crowding_distance(F, front)
        order = np.argsort(-distances)
        slots = pop_size - len(selected)
        selected.extend([front[i] for i in order[:slots]])
        break

    return X[np.array(selected, dtype=int)], F[np.array(selected, dtype=int)]


def tournament_selection(indices, crowding_vals, pool_size=2):
    """Binary tournament selection from a pool of indices based on crowding distance."""
    if len(indices) == 0:
        return 0
    candidates = np.random.choice(indices, size=pool_size, replace=True)
    best = candidates[0]
    for c in candidates[1:]:
        if crowding_vals[c] > crowding_vals[best]:
```

```
            best = c
    return best


def sbx_crossover(p1, p2, lower, upper, prob=0.9, eta_c=15):
    """Simulated Binary Crossover (SBX) between two parents."""
    child = p1.copy()
    if np.random.rand() > prob:
        return child
    for j in range(len(p1)):
        if np.random.rand() > 0.5:
            continue
        if abs(p1[j] - p2[j]) < 1e-14:
            continue
        y1, y2 = min(p1[j], p2[j]), max(p1[j], p2[j])
        yl, yu = lower[j], upper[j]
        rand = np.random.rand()
        # Child 1
        beta = 1.0 + (2.0 * (y1 - yl) / (y2 - y1 + 1e-14))
        alpha = 2.0 - beta ** (-(eta_c + 1.0))
        if rand <= 1.0 / alpha:
            betaq = (rand * alpha) ** (1.0 / (eta_c + 1.0))
        else:
            betaq = (1.0 / (2.0 - rand * alpha)) ** (1.0 / (eta_c + 1.0))
        child[j] = 0.5 * ((y1 + y2) - betaq * (y2 - y1))
        child[j] = np.clip(child[j], yl, yu)
    return child


def polynomial_mutation(x, lower, upper, prob, eta_m=20):
    """Polynomial mutation for a single individual."""
    child = x.copy()
    for j in range(len(x)):
        if np.random.rand() > prob:
            continue
        y = child[j]
        yl, yu = lower[j], upper[j]
        delta = yu - yl
        if delta < 1e-14:
            continue
        r = np.random.rand()
        if r < 0.5:
            deltaq = (2.0 * r) ** (1.0 / (eta_m + 1.0)) - 1.0
        else:
            deltaq = 1.0 - (2.0 * (1.0 - r)) ** (1.0 / (eta_m + 1.0))
        child[j] = y + deltaq * delta
        child[j] = np.clip(child[j], yl, yu)
    return child


def de_rand_1_bin(target, population, front0_indices, lower, upper,
                  F_scale=0.5, CR=0.9):
    """
    DE/rand/1/bin     Differential Evolution operator.

    Select 3 distinct peers from Front 0, create a mutant via
    v = r1 + F*(r2 - r3), then perform binomial crossover with the target.
    """
    n_var = len(target)

    # Build a pool of at least 3 distinct indices
    pool = front0_indices if len(front0_indices) >= 3 else np.arange(len(population))
    chosen = np.random.choice(pool, size=3, replace=False)
    r1, r2, r3 = population[chosen[0]], population[chosen[1]], population[chosen[2]]

    # Mutant vector: v = r1 + F * (r2 - r3)
    mutant = r1 + F_scale * (r2 - r3)
    mutant = np.clip(mutant, lower, upper)

    # Binomial crossover
    j_rand = np.random.randint(n_var)
    child = target.copy()
    for j in range(n_var):
```

```python
        if np.random.rand() < CR or j == j_rand:
            child[j] = mutant[j]

    return np.clip(child, lower, upper)


# ====================================================================
# Main Algorithm
# ====================================================================

def run_mopga(problem, pop_size, generations, seed=1):
    """
    Run the Universal Multi-Objective Phototropic Growth Algorithm.

    Returns the non-dominated Pareto front objective values (F).

    Phase Engine:
        - Cool-down (last 15 generations): pure vector math smoothing
        - Growth Phase (generation % 5 != 0): continuous vectors +
          Island-Aware Magnetic Repulsion
        - Pollination Phase (generation % 5 == 0, not in cool-down):
          Adaptive Multi-Operator Ensemble
            Path A (50%): SBX + M-Dimensional Anchor     orthogonal spacing
            Path B (50%): DE/rand/1/bin, no anchor      rotational exploration
    """
    np.random.seed(seed)
    lower = np.asarray(problem.xl, dtype=float)
    upper = np.asarray(problem.xu, dtype=float)
    n_var = problem.n_var
    n_obj = problem.n_obj

    X, F = initialize_population(pop_size, problem)

    for generation in range(generations):
        fronts, ranks = fast_non_dominated_sort(F)
        crowding_vals = np.zeros(len(F), dtype=float)
        for front in fronts:
            distances = crowding_distance(F, front)
            crowding_vals[np.array(front, dtype=int)] = distances

        offspring = np.zeros_like(X)

        if generation % 5 != 0 or generation >= generations - 15:
            # ===== Growth Phase / Cool-down: phototropic vector math =====
            alpha = np.exp(-generation / max(1, generations))
            beta = 0.4
            auxin = 0.1 + 0.8 * (1.0 / (1.0 + ranks))

            light_mask, shade_mask = divide_light_shade(ranks, crowding_vals)

            # Leaders from the current generation's Front 0
            leaders = (
                X[np.array(fronts[0], dtype=int)]
                if len(fronts) > 0 and len(fronts[0]) > 0
                else X
            )

            def best_group(mask):
                indices = np.where(mask)[0]
                if len(indices) == 0:
                    return None
                best_idx = indices[
                    np.lexsort((-crowding_vals[indices], ranks[indices]))
                ][0]
                return X[best_idx]

            light_best = best_group(light_mask)
            shade_best = best_group(shade_mask)
            if light_best is None:
                light_best = X[np.argmin(ranks)]
            if shade_best is None:
                shade_best = X[np.argmin(ranks)]

```

```
            for i in range(len(X)):
                Xi = X[i]
                leader = leaders[np.random.randint(len(leaders))]

                # Base phototropic force of competition toward the leader
                foc = alpha * beta * np.random.rand(X.shape[1]) * (leader - Xi)

                # Mutate the current solution first for exploration
                strength = 0.05 if light_mask[i] else 0.12
                mutated = mutate_solution(Xi, lower, upper, strength)

                # Directed growth toward the chosen leader weighted by auxin
                directed = auxin[i] * (leader - mutated)

                # Light and shade use different exploitation/exploration patterns
                if light_mask[i]:
                    local_population = X[light_mask]
                    light_move = (
                        alpha
                        * beta
                        * np.random.rand(X.shape[1])
                        * (light_best - mutated)
                    )
                    update = directed + light_move
                else:
                    local_population = (
                        X[shade_mask] if np.any(shade_mask) else X
                    )
                    random_peer = X[np.random.randint(len(X))]
                    shade_move = (
                        alpha
                        * beta
                        * np.random.rand(X.shape[1])
                        * (random_peer - mutated)
                    )
                    shade_best_move = 0.2 * alpha * (shade_best - mutated)
                    update = directed + shade_move + shade_best_move

                # Island-Aware Magnetic Repulsion
                vicinity_candidate = cell_vicinity(
                    mutated + update, local_population, alpha * 0.1, lower, upper
                )
                vicinity_term = vicinity_candidate - mutated

                # Final phototropic growth update
                child = Xi + foc + update + vicinity_term
                offspring[i] = np.minimum(np.maximum(child, lower), upper)

        else:
            # ===== Pollination Phase: Adaptive Multi-Operator Ensemble =====
            mut_prob = 1.0 / n_var
            front0_indices = np.array(fronts[0], dtype=int)

            # M-dimensional anchor: number of position variables to preserve
            num_pos_vars = max(n_obj - 1, 1)

            for i in range(len(X)):
                Xi = X[i]

                if np.random.rand() < 0.5:
                    # --- Path A: Orthogonal Specialist (SBX + M-D Anchor) ---
                    p1_idx = tournament_selection(front0_indices, crowding_vals)
                    p2_idx = tournament_selection(front0_indices, crowding_vals)
                    child = sbx_crossover(
                        X[p1_idx], X[p2_idx], lower, upper, prob=0.9, eta_c=15.0
                    )
                    child = polynomial_mutation(
                        child, lower, upper, prob=mut_prob, eta_m=20.0
                    )
                    # Critical M-Dimensional Anchor: force the child to inherit
                    # the first (n_obj - 1) position variables from its parent
                    # to guarantee perfect spacing on ZDT and DTLZ.
                    for k in range(num_pos_vars):
```

```
                        child[k] = Xi[k]
                else:
                    # --- Path B: Rotational Explorer (DE/rand/1/bin) ---
                    child = de_rand_1_bin(
                        Xi, X, front0_indices, lower, upper,
                        F_scale=0.5, CR=0.9,
                    )
                    # Polynomial micro-mutation for fine-grained exploration
                    child = polynomial_mutation(
                        child, lower, upper, prob=mut_prob * 0.5, eta_m=30.0
                    )
                    # NO anchor here     allows diagonal exploration to slice
                    # through the rotated epistasis traps of the UF suite.

                child = np.minimum(np.maximum(child, lower), upper)
                offspring[i] = child

        F_offspring = evaluate_population(problem, offspring)
        X = np.vstack([X, offspring])
        F = np.vstack([F, F_offspring])
        X, F = select_next_population(X, F, pop_size)

        if generation % 50 == 0:
            print(f"MOPGA generation {generation + 1}/{generations}")

    pareto_front_indices = fast_non_dominated_sort(F)[0][0]
    return F[np.array(pareto_front_indices, dtype=int)]
```